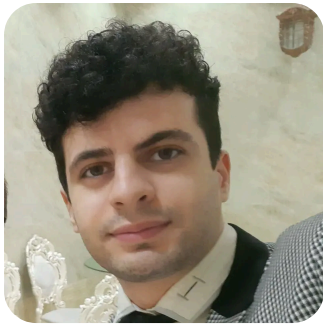




Omid Mosalmani

Senior Backend Developer | AI & LLM Engineer

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GitHub: </OmidMsl>

Website: mark-freeman-personal-website.com

Senior Backend Developer with **5+ years of experience** designing and building **production backend systems**. My primary focus is developing reliable platforms that combine **IoT**, **monitoring** and **alerting**, **automation**, and **AI-powered services**. I have hands-on experience integrating **LLMs** and **Retrieval-Augmented Generation (RAG)** into production applications, supported by a Master's degree in **Artificial Intelligence**.

Skills

Backend

python 5/5 Django 5/5 Django REST Framework 5/5 Rest API 4/5 MQTT 4/5

Authentication 4/5

AI & LLM

LLM Applications 5/5 RAG 4/5 Prompt Engineering 5/5 Fine-tuning 5/5

Model Evaluation 3/5 Hugging Face 3/5 Google Colab 5/5

Frontend

Flutter 4/5 Angular 3/5 TypeScript 2/5 HTML 3/5

DevOps & Infrastructure

Docker 5/5 Nginx 4/5 GitLab 4/5 Linux 4/5 Bash 4/5 Prometheus 4/5

Grafana 4/5 Cloudflare Workers 3/5

Databases

MySQL 5/5 InfluxDB 5/5

Integrations

Firebase 4/5 Telegram Bot API 3/5 Bale Messenger API 5/5 Roboflow 4/5

n8n 3/5

I hereby give consent for my personal data included in my application to be processed for the purposes of the recruitment process.

I speak

Persian

native

English

C1

Work experience

Senior Backend / AI Engineer

January 2024 - now

Setare Soheil Co.

Responsibilities:

- Designed and developed the backend architecture for a large-scale **IoT** greenhouse management platform using Django.
- Designed reliable communication systems for hundreds of IoT devices using **MQTT**, including custom messaging protocols, **fail-safe** mechanisms, and offline **recovery strategies**.
- Built highly available **monitoring** and **alerting** infrastructure using **Prometheus**, **Grafana** and custom health monitoring services.
- Developed **AI-powered features** including **Retrieval-Augmented Generation (RAG)**, **LLM integrations**, **prompt engineering**, **fine-tuning** pipelines, and intelligent greenhouse assistant capabilities.
- Integrated numerous external services including Telegram, Bale Messenger, Roboflow, weather providers, Google Apps Script and Cloudflare Workers to improve **reliability under unstable internet** conditions.
- Managed production infrastructure including **Docker**, **Nginx**, **GitLab**, **Jira**, **Bash automation**, FTP services and **deployment pipelines**.
- Optimized storage architecture using **MySQL** and **InfluxDB** for transactional and time-series sensor data.

Technologies:

Backend: Django, Django REST Framework, MQTT

AI: LLM Applications, RAG, Prompt Engineering, Model Evaluation, Hugging Face

DevOps: Docker, Nginx, GitLab, CI/CD, Prometheus, Grafana, Cloudflare Workers

Databases: MySQL, InfluxDB

Mid-Level Full Stack Developer

June 2021 - January 2024

Setare Soheil Co.

- Developed both frontend (Angular) and backend (Django) for an industrial IoT greenhouse monitoring platform.

- Containerized services using Docker and deployed production environments on Linux servers.
- Designed REST APIs and frontend dashboards for monitoring sensors and greenhouse controllers.
- Collaborated closely with embedded and hardware teams for IoT integration.

Technologies:

Backend: Django, MySQL, Rest API

Frontend & DevOps: Angular, Docker, Linux

Mid-Level Flutter Developer (Freelance)

March 2020 - December 2021

Parscoders, Ponisha

- Developed multiple Flutter applications for clients across different industries.
- Integrated Firebase services including Authentication, Cloud Messaging and Cloud Firestore.
- Implemented Google Maps, location-based services and real-time tracking.
- Developed Bluetooth communication modules for connecting and exchanging data with external hardware devices.
- Delivered applications from design through deployment while working directly with clients.

Technologies:

Mobile: Flutter, Dart

Services: Firebase, Google Maps, Bluetooth Low Energy (BLE)

Junior Android Developer (Freelance)

October 2018 - March 2020

Parscoders, Ponisha

- Developed Android applications in Java with a focus on Augmented Reality features.
- Built foundational experience in Android application architecture and mobile development.

Projects

FarmTech

June 2021 - now

Designed and developed a production IoT platform for greenhouse monitoring and automation. Started as a full-stack developer building the Angular frontend and Django backend, then transitioned to leading backend architecture and AI-powered features.

Designed reliable MQTT communication protocols, offline data synchronization, fail-safe recovery mechanisms, and multi-channel alerting to ensure stable operation despite unreliable internet and power conditions. Built advanced automation features including rule-based

controllers, configurable fuzzy logic, recurring scheduling, and role-based interfaces tailored for users with different technical expertise.

Integrated modern AI capabilities including Retrieval-Augmented Generation (RAG), greenhouse-aware AI assistants, crop disease detection using Roboflow and LLMs, contextual recommendations, and intelligent application support. Managed production infrastructure using Docker, Nginx, Prometheus, Grafana, GitLab, Bash automation, and optimized MySQL and InfluxDB for large-scale sensor and operational data.

Website: <https://app.farmtech.ir/>

Technologies: Django, Angular, MQTT, Docker, Nginx, MySQL, InfluxDB, Prometheus, Grafana, LLM Applications, RAG, Roboflow, Cloudflare Workers, Bale Messenger API, n8n

Persian Leitner

August 2020 - October 2020

Developed a Persian flashcard application based on the Leitner learning method to improve vocabulary retention through spaced repetition. Designed an intuitive user experience for creating, organizing, and reviewing study cards while providing learning statistics and progress tracking.

Download: https://cafebazaar.ir/app/com.omidmsl.english_leitner_box

Repository: https://github.com/OmidMsl/english_leitner_box

Technologies: Flutter, Dart, SQLite

IranGard

August 2021 - October 2021

Developed a tourism application introducing attractions, historical places, and travel destinations across Iran. Implemented interactive maps, location-based features, and responsive mobile interfaces to provide users with an easy way to discover nearby places of interest.

Repository: https://github.com/OmidMsl/iran_gard

Technologies: Flutter, Google Maps, Firebase

SPEC

October 2020 - November 2020

Developed a Flutter mobile application for controlling an industrial air conditioning system through Bluetooth Low Energy (BLE). Built the user interface and device communication flow to allow users to connect to the controller, send commands, and manage AC settings directly from their mobile devices.

Technologies: Flutter, Dart, Bluetooth Low Energy (BLE)

MA App

February 2021 - June 2021

Developed a cross-platform Flutter application for an online marketplace focused on ceramic tiles and related building materials, where sellers were also the producers. Implemented product browsing, ordering workflows, and backend integration to provide a smooth shopping experience for customers and a practical management interface for vendors.

Team size: 2 people

My role: Flutter Developer

Category: Mobile Application, E-commerce

Technologies: Flutter, Firebase, Google Maps, Rest API

Researches

Design and Implementation of an AI Explanation System for Self-Driving Vehicle Decisions

January 2024 - October 2025

M.Sc. Thesis

Designed and implemented a hybrid Explainable AI (XAI) framework for autonomous driving by combining the ADAPT decision-making model with Large Language Models (Gemini and GPT families). The system generates concise, human-understandable explanations using driving decisions, visual attention maps, and prompt engineering techniques.

The research investigated prompt engineering, fine-tuning, multimodal reasoning, and visual grounding using the BDD-X dataset. Experimental results demonstrated significant improvements over the baseline model, achieving up to 11% higher CIDEr-D and 28% higher METEOR scores while producing explanations that were both more fluent and better aligned with the actual driving decisions.

Institution: Shahid Chamran University of Ahvaz

Degree: M.Sc. in Artificial Intelligence

Status: Completed

Keywords: Explainable AI (XAI), Large Language Models, Autonomous Driving, Computer Vision, Prompt Engineering, Fine-tuning, Multimodal AI

An LLM-Based Approach for Clarifying the Decisions of Vision Models in Autonomous Vehicles

September 2025 - November 2025

Conference Paper

Proposed a hybrid framework that combines the ADAPT vision model with Large Language Models to generate trustworthy natural-language explanations for autonomous driving

decisions. The approach bridges visual attention maps and LLM reasoning, improving both explainability and user interpretability in safety-critical systems.

The study evaluated multiple Gemini and GPT models, comparing prompt engineering and fine-tuning strategies using the BDD-X benchmark. Results showed that the fine-tuned Gemini 2.5 Pro model substantially outperformed the baseline across semantic evaluation metrics, with improvements of approximately 28% in METEOR and 11% in CIDEr-D.

Conference: 2025 16th International Conference on Information and Knowledge Technology (IKT) (IEEE)

Keywords: LLMs, Explainable AI, Autonomous Vehicles, Prompt Engineering, Fine-tuning, Vision-Language Models

Education

Artificial Intelligence and Robotics
Shahid Chamran University of Ahvaz
Master degree

September 2022 - September 2025

Computer Engineering
Hamedan University of Technology
Bachelor's degree. Specialization in AI

September 2017 - May 2021